

Counterfactuals and their temporal asymmetries

PHIL UA-78 Metaphysics

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Carroll's 'Mirror Argument'

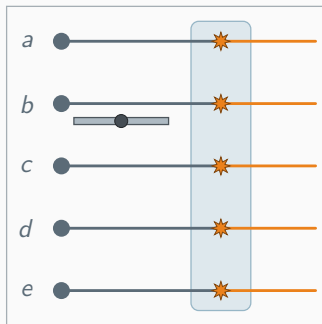
The setup

L *Every X particle that enters a Y -field gets “spin up”.*

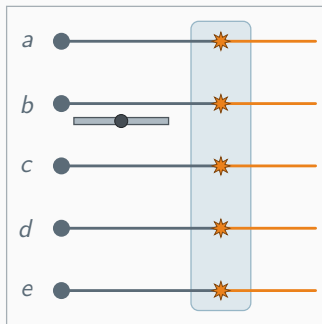
World U_1 5 X -particles a – e that enter 5 Y -fields; each gets “spin up”; mirror not in the way; L is a law.

World U_2 5 X -particles that enter 5 Y -fields; b doesn't get “spin up” though the rest do; mirror not in the way; L is not a law.

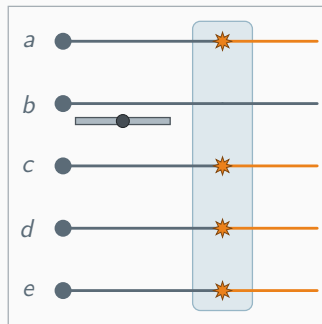
History H Starts off the same as U_1 and U_2 but the mirror is swivelled into the path of b , so it is deflected and never enters its Y -field.



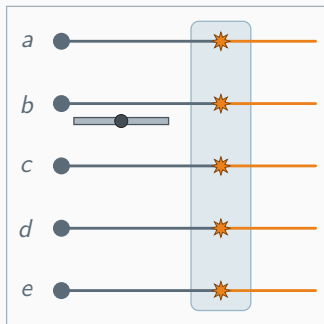
World U_1



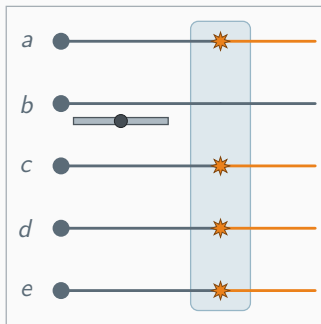
World U_1



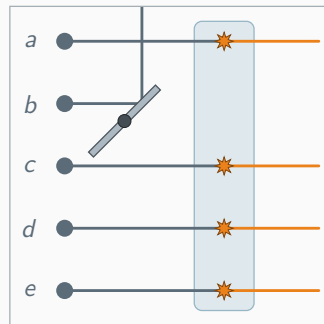
World U_2



World U_1



World U_2



History H

How is this relevant to Humeanism?

If Humeanism is true, then History H is a *complete description* of a possible situation, so, H and L is a law and H and L is not a law cannot *both* be metaphysically possible.

- ▶ Given the Best System Analysis, it's presumably H and L is not a law that's the impossible one. After all, L is simple enough to be a law given the history of U_1 —why would History H be any different?
- ▶ Note however that a version of Carroll's argument will apply whichever way we go on this.

A purely modal argument against Humeanism

1. At U_1 , H is nomically possible and L is a law is nomically necessary.
2. So at U_1 , (H and L is a law) is nomically possible.
3. So, (H and L is a law) is metaphysically possible.
4. At U_2 , H is nomically possible and L is not a law is nomically necessary.
5. So at U_2 , (H and L is not a law) is nomically possible.
6. So (H and L is not a law) is metaphysically possible.

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4. At U_2 , H is nomically possible and L is not a law is nomically necessary.
5. So at U_2 , (H and L is not a law) is nomically possible.
6. So (H and L is not a law) is metaphysically possible.

The BSA-proponent will reject premise 4: at U_2 L is not a law is not nomically necessary.

- It is same-laws necessary, but at U_2 H is not same-laws possible, so premise 4 also fails if we replace 'nomically' throughout with 'same-laws'.

Introduction to counterfactuals

Counterfactual versus indicative conditionals

There's an important gulf between two kinds of interpretations of conditional sentences, marked in many languages by special syntax such as unexpected uses of the past tense. The contrast is illustrated by this famous minimal pair, due to Ernest Adams:

Indicative If Oswald didn't kill Kennedy, no-one did.

Counterfactual If Oswald hadn't killed Kennedy, no-one would have.

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- Note that contrary to what the name may suggest 'counterfactual' does not mean 'has a false antecedent'. In fact, $P > Q$ does not entail, and sometimes does not even *suggest*, that P is false.

If he had that disease, he would be showing exactly these symptoms. (Anderson)

Counterfactuals are not material conditionals

The *material conditional* $P \rightarrow Q$ is equivalent to *Either not- P or Q* .

A few people think that this is a correct semantic analysis of natural language *indicative* conditionals. But AFAIK no-one still thinks it's a correct analysis of counterfactuals.

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That would imply that the 'paradoxes of material conditionals' are valid forms of argument for counterfactuals.

- ▶ Q . Therefore, if P it would be that Q .
- ▶ Not- P . Therefore, if P it would be that Q .

Counterfactuals are context sensitive

Counterfactuals can mean different things in different contexts, as witness these pairs:

If Caesar had fought in the Korean War, he would have used the atom bomb.

If Caesar had fought in the Korean War, he would have used catapults. (Quine)

If I had jumped out of that tenth-floor window, I would have been killed.

If I had jumped out of that tenth-floor window, that would have to have been because someone had a safety net in place. (Jackson)

Are counterfactuals strict conditionals?

Given any notion of necessity $\Box$, the corresponding *strict conditional* is the relation that P bears to Q when $P \rightarrow Q$ is $\Box$ -necessary.

Equivalently: the strict conditional 'if P then Q ' is true when all $\Box$ -accessible P -worlds are Q -worlds.

Some have suggested that counterfactuals are strict conditionals, with the operative notion of necessity varying between different contexts.

Problems for the strict conditional theory

The strict conditional account predicts that the following argument-forms are valid for counterfactuals.

Monotonicity

$$P > Q$$

$$(P \wedge R) > Q$$

Transitivity

$$P > Q$$

$$Q > R$$

$$P > R$$

Contraposition

$$P > Q$$

$$\neg Q > \neg P$$

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Monotonicity	Transitivity	Contraposition
$P > Q$	$P > Q$	$P > Q$
$(P \wedge R) > Q$	$Q > R$	$\neg Q > \neg P$
	$P > R$	

These do not look like great predictions, although they are hard to evaluate because one can always appeal to *context-shift* to explain away seeming counterexamples.

Stalnaker (1968) and Lewis (1973) independently came up with closely related variant approaches using 'closeness' relations on possible worlds (perhaps in addition to an accessibility relation).

Stalnaker's semantics

$P > Q$ is true iff either there is no accessible P -world, or the closest accessible P -world is a Q world.

Lewis's semantics

$P > Q$ is true iff either there is no accessible P -world, or there is an accessible P -and- Q world such that no accessible P -and-not- Q world is closer than it.

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What do 'closer than' and 'accessible' mean? Different things in different contexts.

Lewis furthermore holds that all the different closeness relations have to do broadly with *similarity*, though different *respects* of similarity matter in different ways in

Conditional Excluded Middle

The key difference between Stalnaker and Lewis concerns the status of the following schema:

Conditional Excluded Middle (CEM)

Either $P > Q$ or $P > \neg Q$

Stalnaker argues that CEM is *valid*, and explains this in his semantics by positing that the closeness relation is always such that there is a *unique closest world* in any set of worlds.

Lewis argues that CEM is *invalid*, and accommodates its failure in his semantics by allowing the closeness relations to generate *ties* and *infinite descending sequences*.

A counterfactual-based version of the Mirror Argument

A counterfactual version of the Mirror Argument

Carroll proposes two principles:

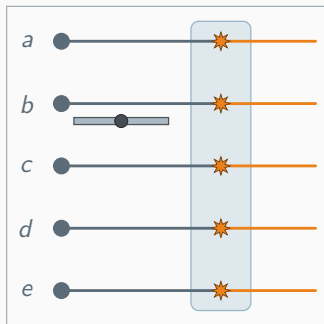
SC*

If P is nomically possible, and L is a law, then if P , L would be a law.

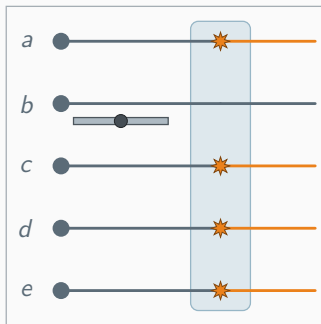
SC'

If P is nomically possible, and L is not a law, then if P , L would not be a law.

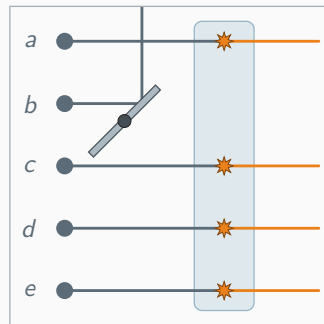
Using these, we get that *If H were true, L would be a law* is true at U_1 , and *If H were true, L would not be a law* is true at U_2 .



World U_1



World U_2



History H

A counterfactual version of the argument

To get from these results to the metaphysical possibility of *H and L is a law* and *H and L is not a law*, we can appeal to the following less-controversial principles:

Counterfactual closure

It is metaphysically necessary that: if the counterfactual *if P then it would be that Q* is true, and *P* is metaphysically possible, then *Q* is metaphysically possible.

Iteration for metaphysical modality

If it's metaphysically possible that it's metaphysically possible that *P*, then it's metaphysically possible that *P*.

Humeans will reject at least one of SC^* and SC' ; best system theorists will reject both.

In the present case, the problem is with SC' : at U_2 , H is nomically possible, and L isn't a law, but if H were true, L would be a law.

- What if we replace 'nomically possible' with 'same-laws possible'? Then it isn't obvious what Humeans should think about SC^* and SC' . But it doesn't matter, since they will deny that H is same-laws possible at U_2 .

From SC to SC* and SC'

Carroll argues for SC* and SC' from the following more basic principle (where $P \rightarrow Q$ stands for the *material conditional*, equivalent to 'either not- P or Q '):

SC

If P is nomically possible, and $P \rightarrow Q$ is nomically necessary, then if P it would be that Q .

Note that given uncontroversial principles of conditional logic, this is equivalent to the following:

SC (variant)

If P is nomically possible, and Q is nomically necessary, then if P it would be that Q .

To apply SC, Carroll has to assume that if L isn't a law, then L *isn't a law* is nomically necessary. But Humeans will reject this.

Is SC true?

SC (variant)

If P is nomically possible, and Q is nomically necessary, then if P it would be that Q .

I don't find it so plausible that this is true in *all* contexts.

If Michelson and Morley had got such-and-such readings, that would have to have been because light was a wave travelling through the ether.

If a baseball were pitched at $0.9c$, everyone in the baseball stadium would soon be killed by a wave of superheated plasma generated by its collision with air molecules. (Holguín and Teitel)

But SC could still be right across some wide range of ordinary contexts.